



A new model for the pressure drop in dual-cell density monoliths: Experimental validation and CFD

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ABSTRACT

This study presents the first experimental validation of a theoretical model for predicting pressure drop in dual-cell density monoliths, a configuration aimed at optimizing flow distribution in industrial catalytic reactors. 3D-printed monoliths, featuring two concentric regions — core and ring — were employed to systematically investigate variations in core size and relative permeability between regions. The monoliths were tested over a wide range of Reynolds numbers using a custom-built experimental rig. The results demonstrate strong agreement with the model's predictions and confirm its accuracy in estimating pressure drop under different configurations. Complementary CFD simulations further support the experimental results and provide additional insights into the effects of model assumptions on pressure drop predictions. The study validates the model as a useful tool for engineering applications, offering valuable insights for the design and optimization of non-homogeneous monolith substrates in industrial catalytic systems.

1. Introduction

Catalytic processes are important to numerous industrial applications, ranging from chemical manufacturing to environmental protection. These processes play a critical role in improving the efficiency and sustainability of chemical plants by optimizing throughput and reducing energy consumption [1]. While packed bed reactors have been the traditional choice, they suffer from high pressure drop, poor flow distribution, and challenges in thermal gradient management [2]. These limitations, especially in large-scale operations, have led to the development of structured reactors, where the geometry, ranging from particle size to reactor scale, is precisely defined by design [3].

Among the structured substrates, monolith reactors have emerged as a promising alternative to traditional packed beds, particularly in high-flow applications. These reactors consist of a structured, honeycomb-like array of parallel channels, which serve as catalyst supports [4]. In packed beds, the pressure drop is typically modeled using the Ergun equation, which accounts for the interactions between the fluid phase and the pellets, leading to both viscous and inertial losses [5] which explains greater pressure drop while increasing flow. However, studies have shown that polydisperse pellet beds, common in industrial reactors, can increase flow resistance, affecting pressure drop, and Ergun predictions underestimate it [6,7]. Moreover, poor flow distribution can create hot spots, particularly in exothermic reactions, as low flow areas concentrate reactants, accelerating conversion rate

and causing localized temperature spikes [8–10]. Maintaining a consistent temperature profile in catalytic reactors is crucial for optimizing efficiency and prolonging catalyst life, as temperature imbalances can degrade catalysts, lower selectivity, and increase undesired by-products formation [11–13].

The use of monoliths as catalytic support offers several improvements, including a significantly lower pressure drop and a higher surface-to-volume ratio compared to packed beds, making them especially suitable for processes requiring high throughput with low pressure drop [14]. Moreover, heat and mass transfer for heterogeneous reactions in monolith reactors is improved by the higher specific area and more even flow distribution, leading to an increased catalyst usage [15,16]. Consequently, monoliths are now widely used in automotive catalytic converters for automobile exhaust treatment and are increasingly considered in various chemical synthesis systems [17,18]; Some examples are CO₂ methanation [19,20], methanol synthesis [21–23], methane reforming [24–26] and destruction of volatile organic compounds (VOC) [27].

Monolith channels can be manufactured in different shapes, such as circular, square, triangular, hexagonal, and sinusoidal, among others [28]. The catalyst is dispersed within the washcoat that is coated onto the inner surface of the channels [29], so the washcoat is a secondary support that provides for better adherence and improved dispersion of the active phase [30]. Some geometric parameters determine the monolith design. In particular, for square-shaped monolith

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channels, there is cell spacing, which refers to the distance from the center of one cell to the center of the adjacent one, and wall thickness (w_i), which denotes the width of the walls separating each channel. Typically, monoliths are also characterized by their cell density (N), which indicates the number of cells per unit cross-section area. It is expressed in cells per square inch (CPSI) [30,31]. All of these parameters together impact the pressure drop and catalytic performance.

Despite the stated benefits, there is still room for improvement in monolith design, particularly in optimizing flow distribution. In catalytic converters, the primary application of monoliths, flow distribution has been extensively studied because the flow that comes from the inlet diffuser tends to concentrate at the center, leading to maldistribution [32]. For an homogeneous monolith with constant cell density, any poor distribution of the incoming flow will tend to persist, as there is no internal characteristic capable of redistributing the flow more efficiently. This results in uncontrolled variations in flow rates and residence times. It has been reported that in such cases, the catalyst within traditional monoliths is only partially utilized, resulting in reduced conversion efficiency and a shortened operational lifespan [33–35]. Important variables, such as pressure drop, are also affected [36,37]. These points could be addressed by manipulating the flow distribution within the monolith geometry. Another factor to consider is temperature control, crucial in monolith applications involving exothermic reactions, which often require external cooling [21]. Despite the manufacturing challenges, highly thermally conductive ceramic and metallic substrates have become commercially available in recent years, which mitigates this restriction. Examples include silicon carbide and titanium aluminate honeycombs used in DPF/SCR filters and corrugated Fe-Cr-Al foil metallic monoliths used in Fischer–Tropsch reactors and high-flux methanation [38]. Recent trials with 3D-printed Ni/SiC monoliths demonstrate thermal stability and low pressure drop in methane reforming at 900 °C [26] supporting the possibility of manufacturing conductive structures on an industrial scale. A viable alternative involves controlling the flow distribution, directing more flow through the outer region of the monolith, convection is enhanced in that area, improving heat transfer and reducing the risk of hot spots and thermal gradients. In that line, the work from Kane and Northrop [39] reports experiments demonstrating improved heat transfer without compromising conversion efficiency in a non-homogeneous concentric substrate, highlighting the advantages of such designs.

Other researchers have also highlighted the potential to manipulate reactor performance using non-homogeneous substrates, particularly dual-cell density monoliths, where the cell density varies radially to form two concentric zones with distinct geometric characteristics [40]. This approach has since been further developed by results from our group [41]. As shown in Fig. 1, the inner zone is referred to as the core, while the outer one is called the ring. The difference in cell density between these zones varies the viscous resistance to flow, resulting in higher flow through the more permeable region. This distribution can be particularly suitable for temperature control in exothermic, externally cooled systems.

Early models for homogeneous monolith reactors, such as those based on the Hagen–Poiseuille equation, assumed fully developed laminar flow within uniform channels and provided reliable predictions for homogeneous monoliths under idealized conditions. Due to computational constraints, an extensively used approach is to model the monolith as a continuous medium. This allows for studying fluid dynamics at a larger scale, eliminating the need to explicitly solve the equations governing phenomena at the channel scale and within the washcoat, which is computationally expensive, even when only a single channel is considered [42,43]. To accurately represent the substrate, the continuum model must include source terms in the governing equations, that emulate the actual phenomena of a fluid passing through physical channels [37,44]. To derive the source term in the momentum equation representing the pressure drop, Darcy's Law is equated with

the Hagen–Poiseuille equation [45], obtaining an apparent permeability for the porous medium that accounts for the energy losses produced in a fluid in channels. Eq. (1) shows this Darcy and Hagen–Poiseuille equivalence for pressure drop $\Delta p/L$.

$$\frac{\Delta P}{L} = \frac{\mu Q}{\alpha_{axial} A} = \frac{C}{2} \frac{\mu u_{ch}}{D_h^2} \quad (1)$$

thus, the apparent permeability is:

$$\alpha_{axial} = \frac{\phi D_h^2}{C/2} \quad (2)$$

while the void fraction is:

$$\phi = \frac{D_h^2}{(D_h + w_i)^2} \quad (3)$$

It can be seen from Eq. (2) that the permeability is determined by the channel hydraulic diameter (D_h), void fraction (ϕ) and a factor that depends on the shape of the channel transversal area (C). These models, while effective for uniform configurations, fail to capture the complexities introduced by dual-cell density monoliths, where the core and ring zones differ in geometrical parameters such as cell density, channel size, or other characteristics to achieve a specific flow distribution. These variations, designed to modulate local flow, result in two distinct permeabilities. Consequently, existing models for homogeneous systems cannot accurately represent the flow dynamics and resistance in these heterogeneous structures, raising the question of how to accurately predict the pressure drop in such substrates.

Recent research utilizing CFD simulation [46] suggests that a back-pressure equilibrium exists between the two concentric zones of the monolith, leading to a theoretical model for predicting the flow distribution in the dual-cell monolith. More specifically, Eq. (4) predicts the fraction of the flow passing through the core region (f_c), as a function of the fraction of the volume of the substrate belonging to the core (λ_c) and the ratio between the apparent permeability of the core (α_c) and the ring (α_r). The latter is referred to as the relative permeability (α_c/α_r). This model assumes fully developed laminar flow.

$$f_c = \frac{\lambda_c}{\lambda_c + (\alpha_r/\alpha_c)(1 - \lambda_c)} \quad (4)$$

Provided the backpressure equilibrium, the pressure drop through a dual-cell density monolith can be computed either by using the core or the ring, as shown in Eq. (5).

$$\frac{\Delta p_r}{L} = \frac{\Delta p_c}{L} = \frac{\mu}{\alpha_c} u_c \quad (5)$$

It is known from previous work that u_c is [46]:

$$u_c = \frac{u_{in}}{\lambda_c + (\alpha_r/\alpha_c)(1 - \lambda_c)} \quad (6)$$

Substituting Eq. (6) into Eq. (5) yields the explicit pressure drop expression for dual-cell density monoliths:

$$\Delta p = \frac{u_{in} \mu L}{\alpha_c \lambda_c + \alpha_r (1 - \lambda_c)} \quad (7)$$

Although this model has been developed to account for the complex flow patterns in non-homogeneous monoliths, it lacks experimental validation. This study fills that gap by experimentally validating the theoretical model for predicting pressure drop in dual-cell density monoliths. In contrast to previous research, which are focused on homogeneous monoliths and factors such as turbulence and entrance/exit effects, this work investigates laminar flow conditions in a single dual-cell monolith. By experimentally validating this theoretical model, this study looks to provide the first comprehensive assessment of its predictive accuracy, adding robustness to a useful tool for accurate pressure drop prediction in these novel systems.

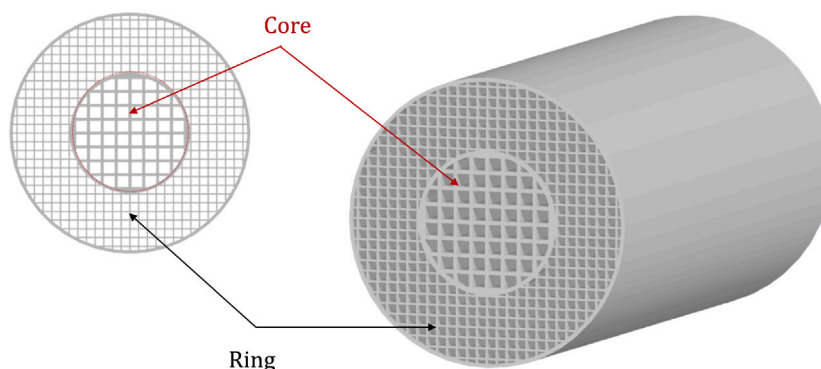


Fig. 1. Monolith with zones of different cell densities.

Table 1

Geometrical parameters of the monoliths used. For all cases $\alpha_c = 1.88 \times 10^{-7} \text{ m}^2$ and $N_c = 42$.

Ring				α_c/α_r	$\lambda_c = A_c/A_r$
D_h , mm	w_i , mm	N , CPSI	α_r , m^2		
2.0	0.45	107	9.39×10^{-8}	2.0	0.25
2.0	0.45	107	9.39×10^{-8}	2.0	0.50
2.5	0.80	59	1.26×10^{-7}	1.5	0.25
2.5	0.80	59	1.26×10^{-7}	1.5	0.50

2. Experimental setup

2.1. Monolith manufacture

Dual-cell density monoliths were digitally modeled using Autodesk Inventor Professional 2023 [47]. For all the cases, the channels were square-shaped due to their ease of design, simplified modeling, and manufacturability through 3D printing, which ensures a more accurate reproduction of geometric parameters. Changes in channel shape would adjust the permeability through the shape factor C , with the model remaining applicable. The core geometric parameters, namely hydraulic diameter and wall thickness, were kept constant across all experiments. That is, the inner zone apparent permeability remained constant. It was calculated by using Eq. (2). Variations were introduced only in the core size, which ranged from 0.25 to 0.50, and in the apparent permeability of the ring zone. As a result, the permeability ratio (α_c/α_r) moved from 1.5 to 2.0. These values correspond to the ones observed in typical monoliths, as axial permeabilities range between 1.2×10^{-8} to $3.1 \times 10^{-8} \text{ m}^2$, [37]. The selected flow area ratios (0.25, 0.50) represent feasible design configurations for dual-zone monoliths, allowing for a representative analysis of flow behavior. Four different monolith configurations were designed for experimentation. Their details are provided in Table 1.

The monoliths were 3D-printed in water-washable resin (Anycubic, grey) with a DLP 3D printer (Creality, Halot-Sky) producing structures 150 mm in length and 68 mm in outer diameter. This material was selected for its ease of fabrication and precision in reproducing complex geometries. Given that the model focuses on flow behavior governed by geometry, monoliths 3D-printed in resin were considered suitable for this study. After printing, the models were washed with water to remove excess resin, air-dried at ambient temperature, and cured under UV light for 20 min. Fig. 2 shows an example of a digital dual-cell density monolith and its 3D-printed counterpart.

2.2. Determination of apparent permeability for the 3D-printed monoliths

The actual hydraulic diameter and wall thickness of the 3D-printed monoliths may differ from the design parameters due to inaccuracies inherent in the manufacturing process. As a result, the apparent

Table 2

Monolith permeability for ring ($D_h=2.0\text{--}2.5$ mm) and core zones ($D_h=3.0$ mm).

D_h , mm	w_i , mm	α design, m^2	ϕ physical	α physical, m^2	Variation, %
2.0	0.45	9.39×10^{-8}	0.57	6.90×10^{-8}	26%
2.5	0.80	1.26×10^{-7}	0.55	1.15×10^{-7}	9%
3.0	0.90	1.88×10^{-7}	0.54	1.60×10^{-7}	15%

permeability of the monoliths may deviate from the intended design. Additionally, although the design specifies a uniform hydraulic diameter, the 3D printing process can introduce slight deviations, resulting in variability in both D_h and w_i across different cells. Those changes are typically minor when using DLP, but still might be present. To address these potential deviations, the lengths of two sets of cells were measured from the physical monoliths using a Vernier caliper. Each set contained a different number of cells, allowing the measurements to capture both the hydraulic diameter and wall thickness. The measurements were then used to set up a system of equations, which was solved to determine the average values of D_h and w_i . This procedure was carried out separately for the core and ring zones of the monolith. To account for possible variations in printing across the structure, these measurements were repeated in multiple directions of the monolith. This dimensional verification showed uniform wall thickness and channel width variations across the monolith, with no systematic defects at core or edges. Surface roughness, while higher due to resin printing, had no impact on flow as laminar conditions prevailed. Printing artifacts were minimized by optimizing resin exposure parameters, as usual in resin 3D-printing.

For each measurement, the apparent permeability was calculated using Eq. (2), and the void fraction was determined using Eq. (3). It should be noted that the presence of washcoat would reduce both the hydraulic diameter and the void fraction, affecting the permeability. However, the model remains applicable if the effective geometry after washcoating is used. These calculations were performed for both the core and ring zones, ensuring accurate characterization of the monolith properties.

Fig. 3 presents the four 3D-printed dual-cell density monoliths used in the experiments, while Table 2 summarizes their key physical characteristics, including measured hydraulic diameters, wall thicknesses, calculated permeabilities, and their variations between that of the digital and the physical monolith.

2.3. Pressure drop experiments

The experimental rig setup is illustrated in Fig. 4, which outlines the arrangement of components used to measure the pressure drop across the monolith substrate. In the setup, a stream of compressed air, controlled by valve V-1, flows through a hose and two parallel flow meters (GFM37, Alborg Instruments). The selection of compressed

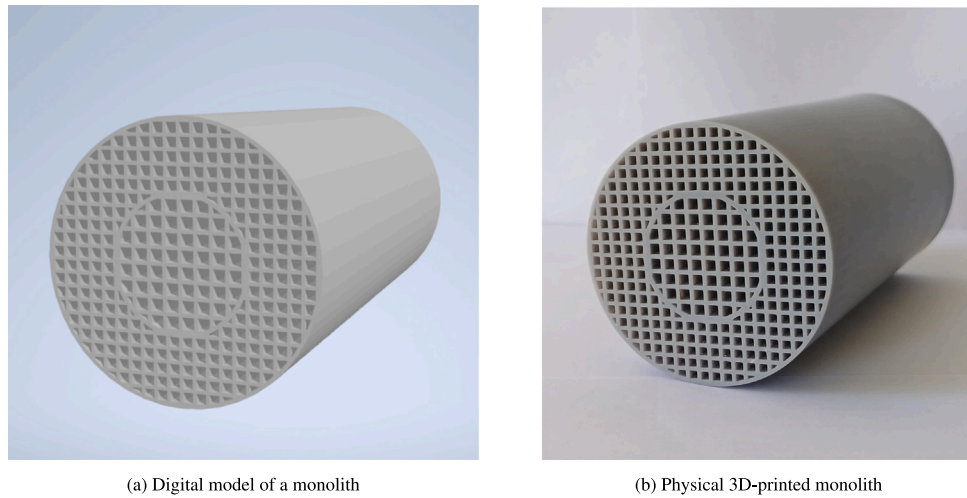


Fig. 2. Design and final structure of a dual-cell density monolith.

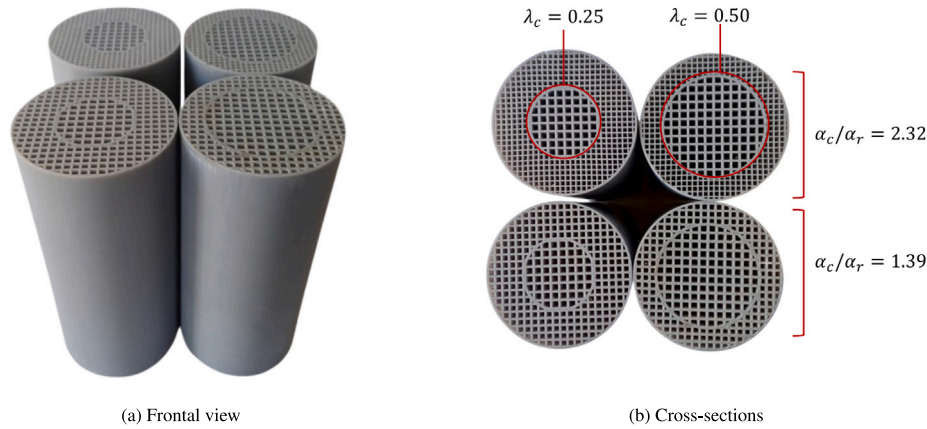


Fig. 3. Four 3D-printed dual-cell density monoliths.

air as the operating fluid was due to its safe handling and well-known properties. The parallel arrangement of flow meters provides accurate measurement across a range of flow rates. The air stream then enters an acrylic pipe, at the end of which the monolith is mounted within a specially designed monolith holder. The monolith discharges directly into the environment, ensuring an open system configuration. The differential pressure transmitter PT-1 (GL42, Ashcroft) was used to measure the pressure drop across the substrate, capturing the difference in pressure before and after the monolith.

For each experimental run, the pressure drop across the substrate, airflow rate, and temperature were recorded to ensure precise monitoring of the system conditions. Ambient temperature was recorded during each run to correct the physical properties of the fluid, namely, density and viscosity. Both the compensated flow (adjusted for temperature variations) and pressure drop were measured in duplicates to ensure statistically representative values. Confidence intervals were determined as $\pm$ one standard deviation from the mean pressure drop value of duplicate measurements and were used to estimate the statistical significance and derive the error bars. The detailed parameters for each experimental run, including flow rates and pressure drops are summarized in Table 3.

3. Computational model

3.1. Description of the domain

The computational domain was made to closely replicate the experimental setup, ensuring an accurate comparison between simulated

Table 3

List of experimental runs. Reynolds number was calculated as $Re_{in} = \rho u D_i / \mu$ and reported values Re_{in} correspond to the average of the duplicates in each run.

n	$\overline{Re}_{in}$	α_c / α_r	$\lambda_c = A_c / A_T$	n	$\overline{Re}_{in}$	α_c / α_r	$\lambda_c = A_c / A_T$
1	85	1.39	0.25	25	133	1.39	0.50
2	211	1.39	0.25	26	218	1.39	0.50
3	315	1.39	0.25	27	322	1.39	0.50
4	427	1.39	0.25	28	440	1.39	0.50
5	538	1.39	0.25	29	540	1.39	0.50
6	638	1.39	0.25	30	635	1.39	0.50
7	749	1.39	0.25	31	753	1.39	0.50
8	856	1.39	0.25	32	860	1.39	0.50
9	956	1.39	0.25	33	954	1.39	0.50
10	1075	1.39	0.25	34	1085	1.39	0.50
11	1173	1.39	0.25	35	1182	1.39	0.50
12	1277	1.39	0.25	36	1293	2.32	0.50
13	83	2.32	0.25	37	84	2.32	0.50
14	210	2.32	0.25	38	208	2.32	0.50
15	317	2.32	0.25	39	312	2.32	0.50
16	430	2.32	0.25	40	431	2.32	0.50
17	536	2.32	0.25	41	539	2.32	0.50
18	631	2.32	0.25	42	649	2.32	0.50
19	746	2.32	0.25	43	755	2.32	0.50
20	847	2.32	0.25	44	862	2.32	0.50
21	946	2.32	0.25	45	963	2.32	0.50
22	1073	2.32	0.25	46	1077	2.32	0.50
23	1167	2.32	0.25	47	1175	2.32	0.50
24	1277	2.32	0.25	48	1283	2.32	0.50

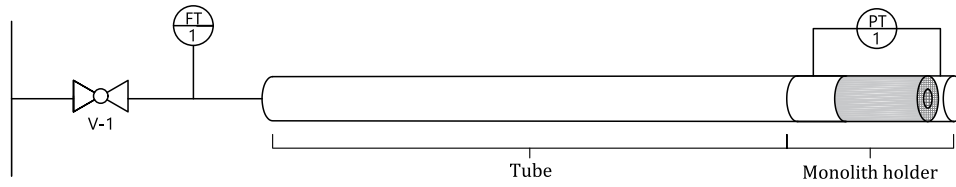
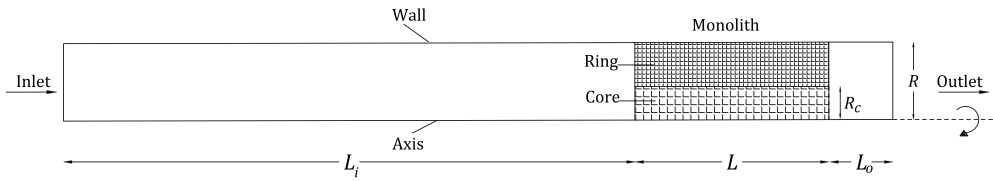


Fig. 4. Experimental rig. The tube length and internal diameter were $L_t=1550$ mm and $D_t = 69$ mm, respectively.



(a) 3D schematic of the domain



(b) 2D computational domain. The 2D generatrix plane rotates around the bottom axis to compose the volumetric domain.

Fig. 5. Computational domain. $L_i=1550$ mm, $L=150$, $L_o=30$ mm, and $R=34.5$ mm.

and laboratory results. The geometry consisted of an axisymmetric tube with a dual-cell density monolith placed within it, as depicted in Fig. 5. The monolith length (L) was 150 mm, with a radius (R) of 34.5 mm. To capture the full development of the flow, two additional sections were included: an upstream section with a length L_i of 1550 mm and a downstream one with a length L_o of 30 mm. These lengths are sufficient to avoid boundary effects.

The monolith itself contained two distinct regions, the core and the ring, which differed in cell density and geometric properties. These structural variations led to differences in apparent permeability. Both regions feature unwashcoated channels with square cross-sections. The permeability variations were introduced to simulate realistic changes in channel diameter and wall thickness between the core and ring sections.

The monolith was modeled as a continuum porous medium. This approach significantly reduced the computational cost by treating the monolith as a homogeneous material where the flow experiences equivalent heat, mass and momentum transfer as it would in discrete channels [48,49]. The permeability in each region was tailored to reflect the dual-cell density structure, capturing the essence of flow distribution without requiring the resolution of individual channel details.

3.2. Flow model

The flow regime at the inlet was assumed to be turbulent, and it was modeled using the Reynolds-Averaged Navier–Stokes (RANS) equations with a two-equation turbulence model. The continuity and momentum conservation equations were applied as described by [50]:

$$\frac{\partial (\rho u_i)}{\partial x_i} = 0 \quad (8)$$

$$\frac{\partial (\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \frac{\partial u_i}{\partial x_i} \delta_{ij} \right] - \frac{\partial \tau_{ij}}{\partial x_j} + S_{u_i} \quad (9)$$

The Boussinesq hypothesis was employed to close the system [51]. The Reynolds stress tensor, τ_{ij} , was modeled as:

$$-\tau_{ij} = \overline{u'_i u'_j} = \mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \left(\frac{\partial u_i}{\partial x_i} \right) \delta_{ij} \quad (10)$$

Turbulence viscosity, μ_t , was calculated using the Shear Stress Transport (SST) model [52], which combines the advantages of both the $k-\epsilon$ and $k-\omega$ models. The turbulence viscosity is expressed as:

$$\mu_t = \rho \frac{k}{\omega} \frac{1}{\text{Max} \left(\frac{1}{\alpha^*}, \frac{S F_2}{a_1 \omega} \right)} \quad (11)$$

The transport equations for the turbulence kinetic energy and specific dissipation rate were [52]:

$$\frac{\partial (\rho k u_i)}{\partial x_j} = \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + \mu_t S^2 - \rho \beta^* k \omega + S_k^{\text{sink}} + S_k^{\text{gen}} \quad (12)$$

$$\frac{\partial (\rho \omega u_i)}{\partial x_j} = \left[\left(\mu + \frac{\mu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial x_j} \right] + \alpha_\omega \alpha^* \rho \mu_t S^2 - \rho \beta^* \omega^2 + 2(1-F_1) \rho \frac{1}{\omega \sigma_{\omega_2}} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j} + S_\omega^{\text{sink}} + S_\omega^{\text{gen}} \quad (13)$$

There is substantial evidence in the literature that once turbulent flow enters the monolith, and there is a sub-critical Reynolds number inside of the channels, then turbulence quickly decays [53]. Based on that, inside of the monolith, both turbulence kinetic energy and specific dissipation ratio were set to zero. Despite that, the flow does not become steady; instead, it becomes laminar unsteady in a sort of pulsatile regime. Such a phenomenon has been previously investigated by our group, reporting that it does not significantly affect the pressure drop [54]. Therefore, assuming a steady laminar flow friction factor still recognizes pulsating flow without violating mass conservation. It is also known that turbulence can arise because of the flow leaving the substrate [55]. However, this has only a minor effect on total pressure drop and can be neglected for most of the cases [56,57].

The monolith itself was modeled as a continuous porous medium, with a source term in the momentum equation representing the extra pressure drop because of the flow inside the channels. Darcy law was used in conjunction with the Hagen–Poiseuille equation to derive the source term, which is valid for fully developed laminar flow. That term appears in the momentum equation as [58,59]:

$$S_{u_i} = -\frac{\mu}{\alpha_i} u_i \quad (14)$$

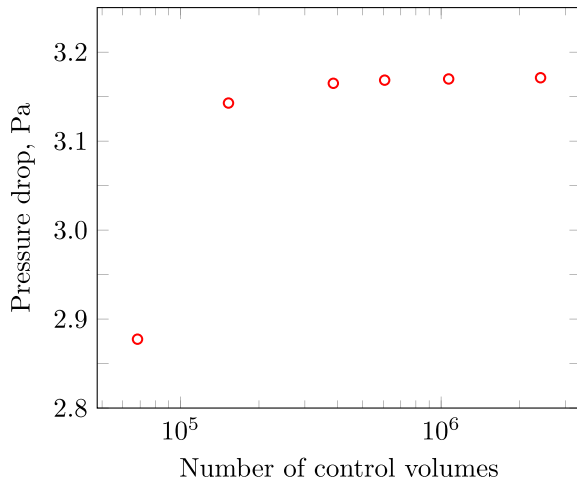


Fig. 6. Pressure drop through the monolith for different grid sizes.

In the monolith, the flow is confined primarily to the axial direction of the channels. Therefore, to correctly represent this in the computational domain, the axial permeability was set to be three orders of magnitude larger than the radial one, as shown below:

$$\alpha_{axial_n} = \frac{\phi_n D_n^2}{C_n/2}; \quad \alpha_{radial_n} = \frac{\alpha_{axial_n}}{1000} \quad (15)$$

In Eq. (15), the n subscript refers to the core ($n = c$) or ring ($n = r$). For the boundary conditions, a velocity inlet was prescribed, which varied according to the Reynolds number for each computational run. At the outlet, a pressure boundary condition of 1 bar was imposed. The top boundary of the domain was set as a solid wall, with no-slip condition in the monolith zone, as the Darcy term accounted for all frictional effects inside the substrate. The bottom boundary represented the axis of rotational symmetry, consistent with the axisymmetric geometry of the domain. The fluid used for all simulations was atmospheric air, with its properties evaluated at the experimentally measured temperature in the respective run. The simulations were performed using ANSYS Fluent 2023 R2 [50], which employs the finite volume method for numerical solutions.

3.3. Grid study and model validation

A grid independence test was performed to ensure the mesh quality and validate that the results were independent of the grid size [60,61]. The pressure drop across the monolith was monitored for six different grid sizes, as this variable is both sensitive to the grid resolution and critical to the study objectives. The grid, fully composed of rectangular elements, was progressively refined, with the number of control volumes ranging from 680 000 to 2 400 000. All simulations were performed under the same conditions, with $Re_{in} = 630$ and $\alpha = 1.15 \times 10^{-7} \text{ m}^2$. The results in Fig. 6, indicate that, beyond the fourth grid size (corresponding to 607 250 control volumes), the difference in the pressure drop is negligible. Therefore, further refinement of the grid would not yield significant improvements, and the study continued with this mesh size.

In addition to the grid study, a validation of the results was conducted. The pressure drop across a monolith with homogeneous permeability (or uniform cell density) was compared against a well-established theoretical prediction—the Hagen–Poiseuille equation. This expression is highly validated for laminar flow through channels and is given by [58,59]:

$$\Delta p = \frac{D_h}{L} \frac{56.8}{Re_{ch}} \frac{1}{2} \rho u_{ch}^2 \quad (16)$$

In the simulation, the pressure drop was determined by calculating the energy loss between the inlet and outlet faces of the monolith

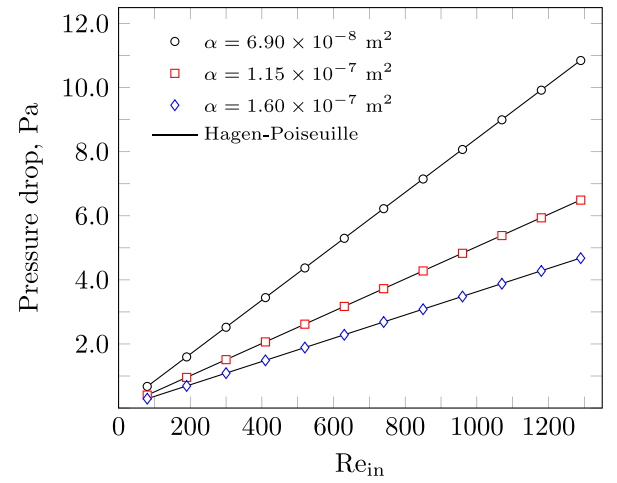


Fig. 7. Comparison between the pressure drop from the computational model and that from Hagen–Poiseuille equation.

Table 4
List of computational runs.

n	Re _{in}	α_c/α_r	$\lambda_c = A_c/A_T$	n	Re _{in}	α_c/α_r	$\lambda_c = A_c/A_T$
1	80	1.39	0.25	25	80	1.39	0.50
2	190	1.39	0.25	26	190	1.39	0.50
3	300	1.39	0.25	27	300	1.39	0.50
4	410	1.39	0.25	28	410	1.39	0.50
5	520	1.39	0.25	29	520	1.39	0.50
6	630	1.39	0.25	30	630	1.39	0.50
7	740	1.39	0.25	31	740	1.39	0.50
8	850	1.39	0.25	32	850	1.39	0.50
9	960	1.39	0.25	33	960	1.39	0.50
10	1070	1.39	0.25	34	1070	1.39	0.50
11	1180	1.39	0.25	35	1180	1.39	0.50
12	1290	1.39	0.25	36	1290	2.32	0.50
13	80	2.32	0.25	37	80	2.32	0.50
14	190	2.32	0.25	38	190	2.32	0.50
15	300	2.32	0.25	39	300	2.32	0.50
16	410	2.32	0.25	40	410	2.32	0.50
17	520	2.32	0.25	41	520	2.32	0.50
18	630	2.32	0.25	42	630	2.32	0.50
19	740	2.32	0.25	43	740	2.32	0.50
20	850	2.32	0.25	44	850	2.32	0.50
21	960	2.32	0.25	45	960	2.32	0.50
22	1070	2.32	0.25	46	1070	2.32	0.50
23	1180	2.32	0.25	47	1180	2.32	0.50
24	1290	2.32	0.25	48	1290	2.32	0.50

substrate. The total pressure at each face was averaged using a mass-weighted approach, as described by Eq. (17). This method ensures consistency with the momentum conservation principle and represents the pressure behavior across the monolith.

$$p|_x = \frac{\int_A p u dA}{\int_A u dA} \quad (17)$$

Fig. 7 presents the pressure drop results for three different permeability values and a range of inlet Reynolds numbers (from 80 to 1290). The computational model shows close agreement with the Hagen–Poiseuille equation across all cases, validating the accuracy of the simulation for predicting pressure drop behavior in monoliths.

The list of computational runs is detailed in Table 4, which shows the variations in core size (ranging from 0.25 to 0.50) and relative permeability values (α_c/α_r) between 1.39 and 2.32. These conditions were selected to match those used in the physical experiments, ensuring that the simulations match real-world behavior.

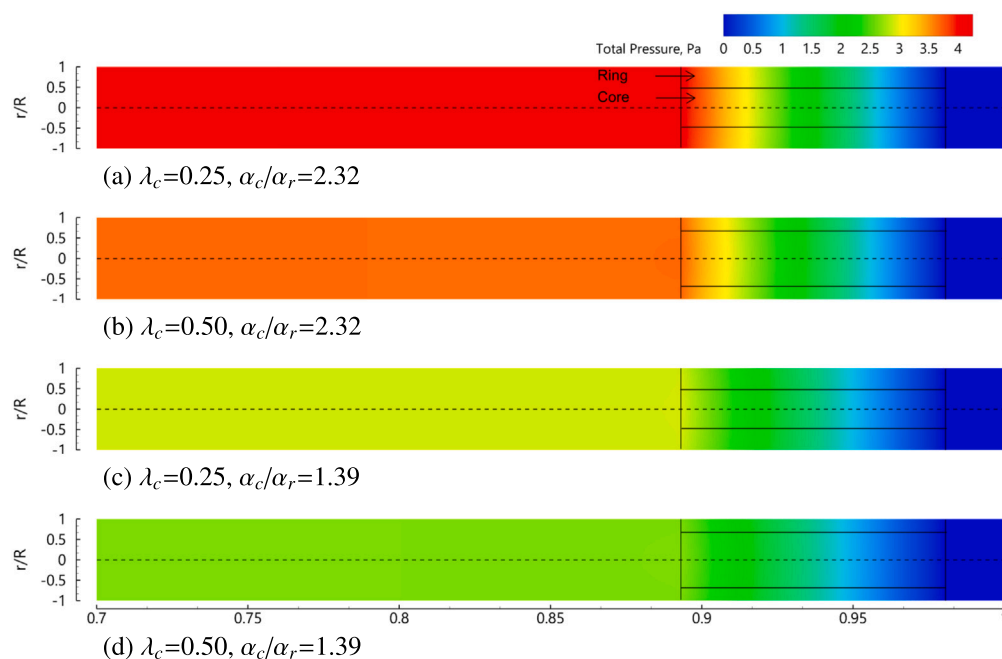


Fig. 8. Pressure drop distribution for different apparent permeabilities and core sizes. The x-axis is scaled as $x/(L_i + L_o)$.

4. Results and discussion

This section presents the main results obtained from the experimental trials and the corresponding simulation runs. The analysis is divided into two parts: pressure drop behavior and flow distribution through the dual-cell density monolith. Both experimental and computational findings are compared to assess the accuracy of the computational model and its potential for practical applications.

4.1. Backpressure and flow distribution

Fig. 8 shows the pressure distribution for the section of the domain close to the substrate for an inlet Reynolds number of 650. The figure covers all the core sizes and apparent permeabilities under study. Notably, in the monolith area, the pressure is mostly uniform in the radial direction across the whole monolith, confirming the backpressure equilibrium assumption between the core and ring zones. In addition, it is observed from comparing Fig. 8(a) and (b) that the pressure drop increases with a smaller core size due to the reduction of the size of the more permeable region. Regarding the effect of the relative permeability, the differences between Fig. 8(a) and (c) show that a higher ratio or disparity leads to a higher pressure drop as well.

Similarly, the velocity profile within the same zone of tube is displayed in Fig. 9. It is observed that in the monolith region, the velocity magnitude profile seems constant at their own core and ring values. At smaller core size values, the variation in flow distribution becomes more significant, increasing the velocity value through the core. Likewise, at larger relative permeabilities, the flow tends to increase through the more permeable region as the difference in resistance to flow between the core and ring becomes more significant.

To further analyze the flow distribution, Fig. 10 reports the axial velocity profile at the outlet of the substrate for a homogeneous monolith ($\alpha_c/\alpha_r = 1.00$). It is compared to that from a dual-cell density monolith with the core region being more permeable ($\alpha_c/\alpha_r = 2.32$) and one with the ring being more permeable ($\alpha_r/\alpha_c = 2.32$), interchanging their local permeabilities. The axial velocity profiles of the dual-cell density monoliths mirror each other symmetrically, indicating that the permeability distribution significantly affects the flow field at the monolith exit. The more permeable region consistently drives more flow, leading to significantly different outlet velocity profiles compared to the homogeneous case.

4.2. Pressure drop model

As previously mentioned, the pressure drop through the monolith, by equals the ring and core, can be modeled by using Eq. (7), which depends on the geometrical parameters of the core and ring, flow rate and the properties of the operating fluid. The equation was tested against experimental and CFD data for different relative permeabilities, core sizes and Re_{in} . The results are summarized in Fig. 11.

The results in Fig. 11 show that the theoretical model closely follows the trends observed in both the experimental and CFD data, particularly at lower Reynolds numbers. This agreement evidences that the model effectively captures the main factors influencing pressure drop, even in configurations with varying core sizes and apparent permeabilities.

For the case of $\lambda_c=0.25$ and $\alpha_c/\alpha_r=1.39$ (Fig. 11(a)), the experimental and CFD results are in strong agreement, and the theoretical model closely matches both datasets. Minor deviations appear when increasing Reynolds numbers, where the theoretical model slightly underestimates the pressure drop. A similar trend is observed for $\lambda_c=0.50$ and $\alpha_c/\alpha_r=2.32$ (Fig. 11(c)), where the theoretical model again performs well, but shows small discrepancies at higher Re. These results suggest that the theoretical model is robust in predicting pressure drop for small-core configurations, with only slight underestimations at higher flow rates.

In the case of a larger core with lower apparent permeability (Fig. 11(b)), the theoretical model also performs well, closely aligning with both the experimental and CFD data across the full range of Reynolds numbers. However, as the apparent permeability increases in the larger-core configuration (Fig. 11(d)), the discrepancies between the theoretical model and the experimental/CFD results become more pronounced at higher Re. While the model still provides a reasonable approximation, it tends to underestimate the pressure drop more visibly as Reynolds numbers increase. The underestimation of the pressure drop in the mentioned cases suggests that additional factors, such as entrance and exit effects or more complex flow resistance mechanisms, may become important in higher flow rate conditions.

All cases show a linear relationship between Reynolds number and pressure drop, confirming that the structure of the proposed theoretical model is fundamentally correct. This linearity supports a core assumption of the model that pressure drop is dominated by fully

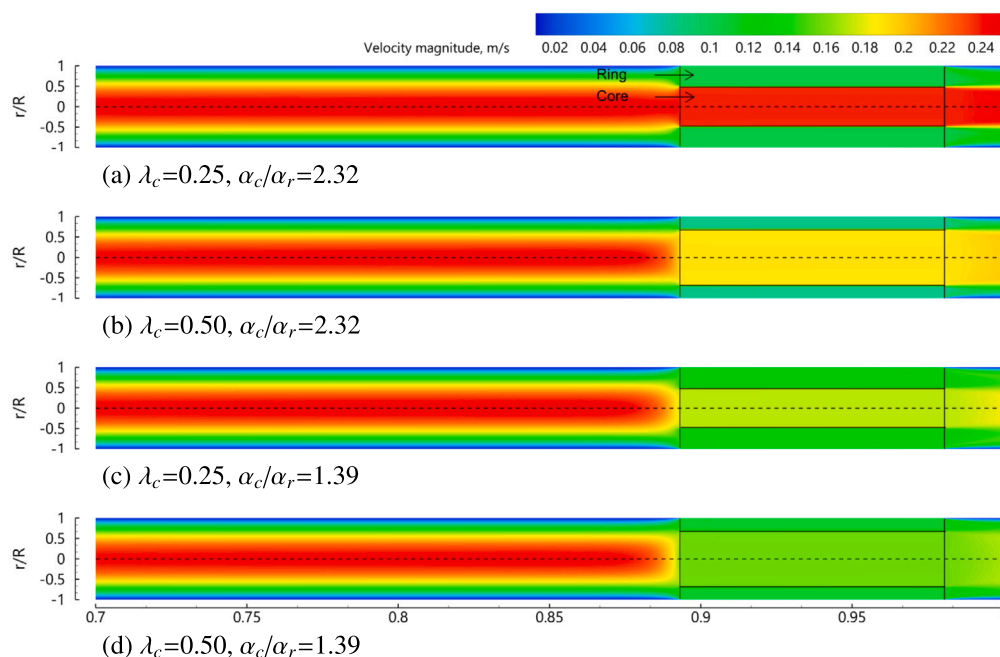


Fig. 9. Velocity distribution for different apparent permeabilities and core sizes. The x -axis is scaled as $x/(L_i + L_o)$.

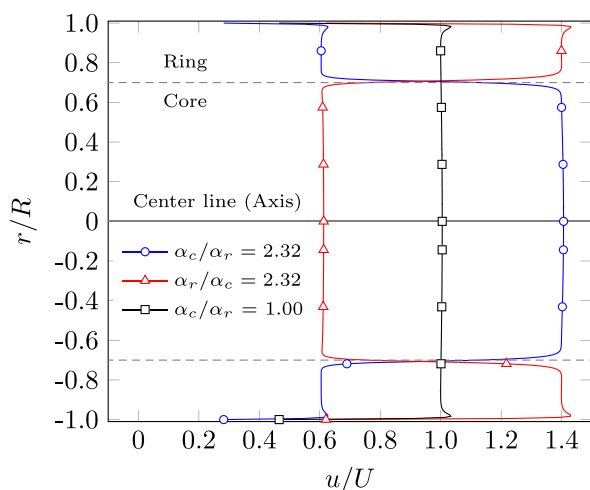


Fig. 10. Velocity profile at the outlet of the monolith for different relative permeabilities (α_c/α_r). U is the area-weighted average velocity.

developed laminar flow inside the parallel channels of the monolith. However, slight variations in the slope between the experimental, CFD, and theoretical results suggest differences in the actual permeability of the physical monolith and the values assumed in both the theoretical and computational models. These discrepancies might be attributed to minor manufacturing imperfections, which are typical in extruded monoliths but less pronounced in 3D-printed models. In this study, DPL (Digital Light Processing) 3D printing was used, which is known for its superior precision compared to traditional extrusion methods. Nevertheless, despite this fact, small manufacturing irregularities in the monolith were still observed. Variations in channel sizes and wall thicknesses, even within the same region, were detected when measuring the physical 3D-printed monoliths. These imperfections may have affected the permeability and contributed to the observed deviations. Additionally, the theoretical and CFD models assumed that all channels were perfectly square. Meanwhile, some real channels are partially truncated by the boundaries of the core region and the shell

of the monolith. Those boundary channels have a smaller hydraulic diameter compared to regular square channels. This leads to a lower local apparent permeability, which in turn increases the slope of the pressure drop versus Reynolds number trend, while maintaining its linear character. This does not affect the backpressure equilibrium assumption; however, it may become more significant as the fraction of channels with non-regular shapes increases.

Overall, the differences between the model and the experiments remained small. The marked alignment between the experimental and CFD results with the theoretical model across all cases suggests the reliability of both the experimental setup and the CFD approach, further validating the proposed model for pressure drop in dual-cell density monoliths.

As an additional analysis, the results presented in Fig. 12 further emphasize the usefulness of the proposed model for predicting pressure drop in dual-cell density monoliths. As shown in both plots, the pressure drop for dual-cell configurations may significantly deviate from that of homogeneous monoliths, whether assuming the permeability of the core (black dashed line) or the ring (red dashed line). This discrepancy is particularly pronounced in cases of high apparent permeability (Fig. 12(b)), where the experimentally measured pressure drop (red squares and black circles) may lie substantially distant from the predictions for homogeneous monoliths. In both low and high apparent relative permeability cases, neither the core nor the ring permeability alone is sufficient to capture the pressure drop behavior of the dual-cell structure, as the pressure drop for dual-cell monoliths results from the complex interaction between the core and ring regions together with the backpressure equilibrium. These findings highlight the limitations of existing models that assume a single permeability and emphasize the proposed model's reliability, specifically designed to account for the non-homogeneous nature of dual-cell density monoliths. Therefore, this model can accurately predict pressure drop in such configurations, which opens opportunities for further monolith shape optimization.

5. Conclusions

This study provides the first experimental validation of a theoretical model designed to predict pressure drop in dual-cell density monoliths. Using 3D-printed monoliths with varying core sizes and relative permeabilities, the experimental results strongly correlate with the model

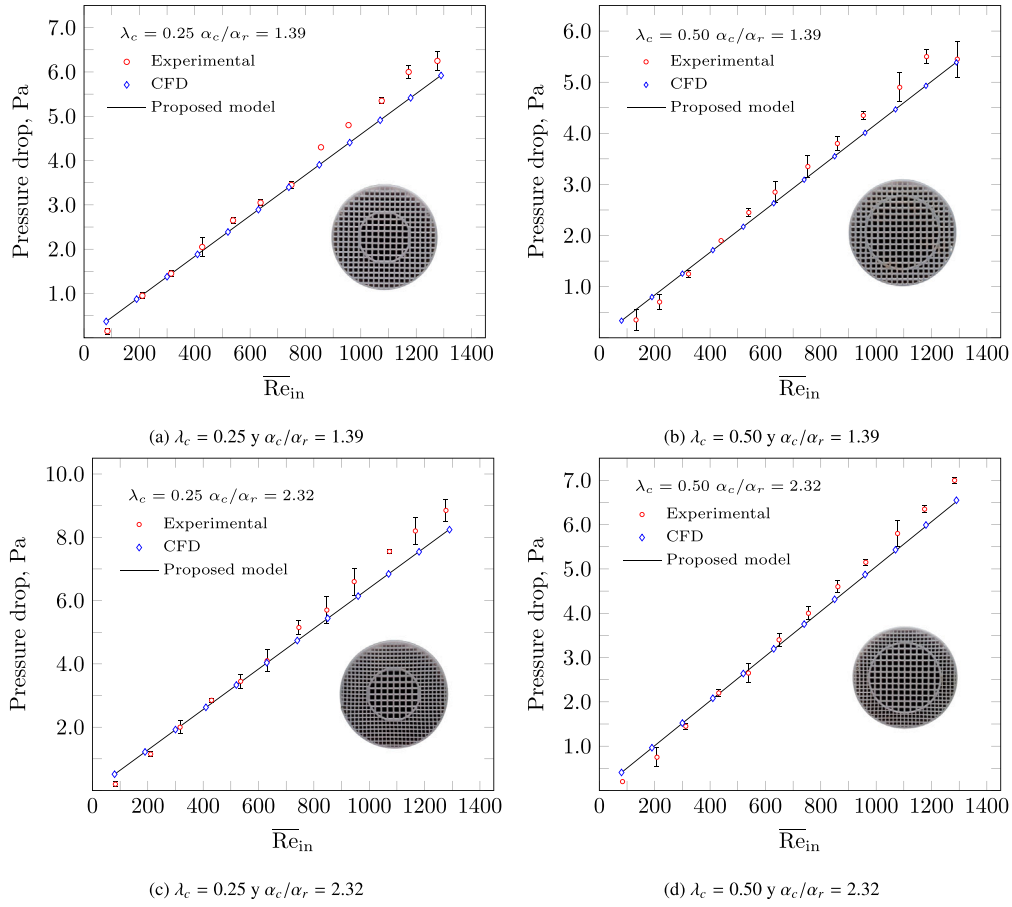


Fig. 11. Pressure drop from the experiments, CFD and theoretical model for different dual-cell density monoliths.

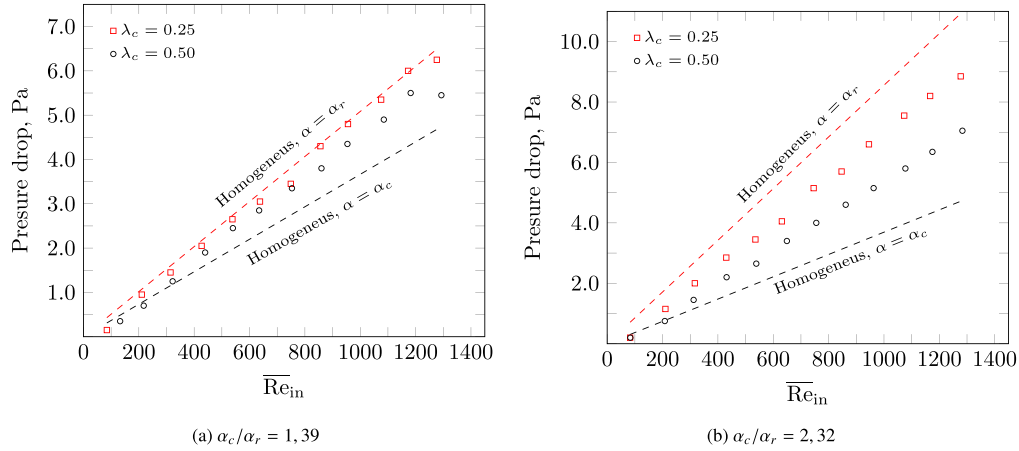


Fig. 12. Comparison of pressure drop between dual-cell density and homogeneous monoliths.

predictions, confirming its robustness across a range of conditions. The model effectively captures the pressure drop behavior of dual-cell density monoliths, where pressure drop increases with higher permeability ratios and smaller core sizes, consistent with the model formulation.

Moreover, the findings also confirm the existence of a backpressure equilibrium between the core and ring zones, supporting a key assumption of the theoretical formulation. Complementary CFD simulations further validate these observations, supporting that the model accurately predicts flow dynamics and resistance in non-homogeneous monolith structures. While minor deviations in pressure drop are observed at higher Reynolds numbers — likely due to entrance and exit

effects — these factors do not significantly impact the overall accuracy of the model.

Compared to homogeneous substrate configurations, dual-cell density monoliths exhibit different pressure drop characteristics, underscoring requiring specialized models like the one validated in this study to predict the behavior of non-homogeneous substrates accurately. The results confirm the reliability of the analyzed model, making it a useful tool for optimizing monolith designs in industrial catalytic applications.

Future work could extend the experimental validation of the model to a broader range of monolith configurations, including, for example, different cell geometries. This would help assess the model robustness

across more diverse applications. Additionally, investigating the impact of entrance and exit effects more thoroughly, especially at higher Reynolds numbers and low void fractions, could provide deeper insights into flow dynamics and further refine the model predictive capabilities. Additionally, optimizing dual-cell density monoliths for specific catalytic reactions, including those involving highly exothermic reactions, could open new opportunities for improving reactor performance.

Nomenclature

A_c	Core cross-sectional area, m ²
A_r	Ring cross-sectional area, m ²
A_T	Total cross-sectional area, m ²
C	Shape factor, -
D_h	Hydraulic diameter, m
D_t	Tube internal diameter, m
f_c	Fraction of flow passing through the core, m
k	Turbulence kinetic energy, m ² /s ²
L	Monolith length, m
L_i	Inlet tube length, m
L_o	Outlet tube length, m
L_t	Tube length, m
N	Cell density, CPSI
p	Pressure, Pa
Δp	Pressure drop, Pa
Δp_c	Core pressure drop, Pa
Δp_r	Ring pressure drop, Pa
R	Total or tube radius, m
R_c	Core radius, m
Re_{ch}	Channel Reynolds number, $= \rho u D_{ch} / \mu$
Re_{in}	Inlet or tube Reynolds number, $= \rho u D_t / \mu$
u	Velocity magnitude, m/s
u_{ch}	Channel velocity, m/s
w_t	Wall thickness, m
α	Apparent permeability, m ²
α_c	Core apparent permeability, m ²
α_r	Ring apparent permeability, m ²
δ_{ij}	Kronecker delta, -
ϕ	Void fraction, -
ρ	Fluid density, kg/m ³
μ	Fluid viscosity, Pa·s
μ_t	Turbulence viscosity, Pa·s
ω	Specific turbulence dissipation rate, 1/s

CRedit authorship contribution statement

Consuelo Reinao: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Pablo Diaz:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis. **Ivan Cornejo:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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